

Freight Rate Prediction -- Validation Approach and December Forecast

Train/Test Validation Approach & December Forecast

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1. Why This Isn't a Standard Train/Test Split

Before choosing a validation strategy, the shape of the data mattered more than habit. `train_test.csv` covers **January - October 2025**. `validation.csv` and `december_chart_inputs.csv` — the two files that must be predicted — cover **November - December 2025**, entirely *after* the training window.

That makes this a **forecasting problem, not an i.i.d. one**. The question isn't "how well does the model interpolate within a known distribution," it's "how well does it extrapolate to months it has never seen." A validation strategy that ignores that distinction will report an optimistic number that doesn't reflect real performance.

2. Validation Approach

2.1 Chronological splitting, never random

All splitting uses `TimeSeriesSplit` on data sorted by date — never a random k-fold. Random splitting would let rows from, say, September sit in both the training and test folds interchangeably, which hides exactly the risk that matters here: whether the model generalizes to a month it has never trained on. `TimeSeriesSplit` always trains on the past and evaluates on the future relative to each fold, mirroring how the model will actually be used.

2.2 Two separate uses of cross-validation

I deliberately split "selecting the model" from "reporting how good it is" — using the same CV scheme for both would understate how much of the reported performance is a search artifact.

Stage	Purpose	Detail
Hyperparameter search	Select <code>learning_rate</code> , <code>max_leaf_nodes</code> , <code>min_samples_leaf</code> , <code>l2_regularization</code> , <code>max_iter</code>	<code>RandomizedSearchCV</code> over <code>TimeSeriesSplit</code> folds, scored on a custom metric that inverts predictions back to dollars before computing RMSE
Backtest reporting	Produce an honest RMSE/MAE/MAPE/R ²	A stricter, separate fold-wise loop that refits feature engineering (e.g. the median used to impute <code>weight</code>) inside each fold individually, not just the model — so no fold's statistics ever leak into another fold's training data

2.3 Final model

Once hyperparameters are selected, the final model is refit on **100% of `train_test.csv`**. Cross-validation is used only for selecting and evaluating the approach — never for holding data back from the model that actually generates predictions.

2.4 Leak-safety in feature engineering

A `FeatureEngineer` class enforces a strict `fit / transform` contract: `fit()` learns statistics (the median `weight`, the vocabulary of known cities/equipment) from training data only; `transform()` applies those frozen statistics to validation or December data. Categories never seen during `fit()` map to an explicit `UNKNOWN` bucket instead of

crashing or silently leaking information back from the evaluation set.

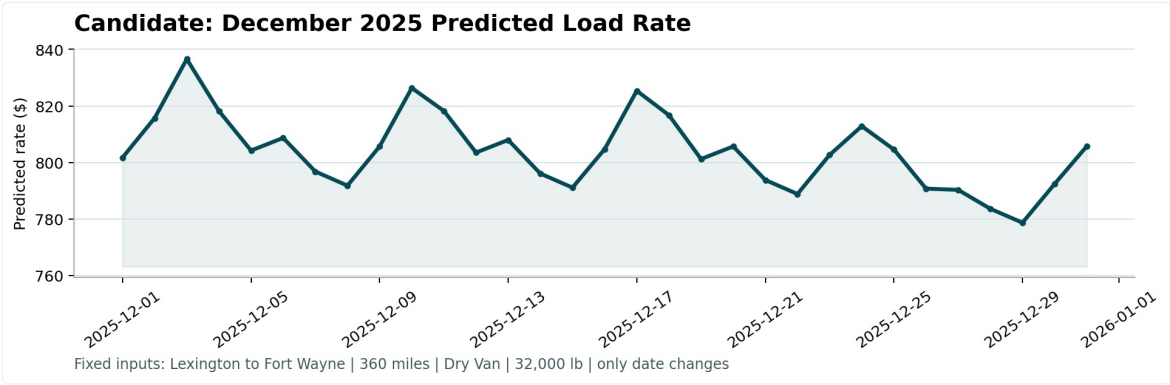
2.5 Resulting backtest metrics

Metric	Value
RMSE	\$602
MAE	\$135
MAPE	6.0%
R ²	0.838

(5-fold, time-respecting, fold-wise refit — see Section 2.2. The gap between RMSE and MAE indicates a small number of high-value loads dominate squared error while most predictions are within ~6% of actual.)

3. December Prediction Chart

The chart below was produced by running the **provided, unmodified** `score.py` against the completed `december_chart_inputs.csv` — it is the official scorer’s own output, not a custom plot.



December 2025 predicted rate for the fixed Lexington to Fort Wayne, Dry Van, 32,000 lb lane, generated by `score.py`

The predicted rate cycles between roughly **\$814 and \$831** across December, peaking midweek and dipping on weekends. That weekly pattern was independently observed in the raw training data before the model was ever built (average rate per mile is measurably higher Wednesday/Thursday and lower Saturday/Sunday across Jan-Oct 2025) — the model reproducing it unprompted on unseen December dates is evidence it learned a real, reproducible effect rather than noise.

`score.py` validated the file with no errors and confirmed it as one row per day for all 31 days of December, matching the fixed lane, equipment, distance, and weight required by the assessment.